

Assignment 3

Course: Machine Learning Foundations

Level: Advanced / Competitive

Instructions

This assignment is designed to test your **Higher-Order Thinking Skills (HOTS)**. You are expected to think like a Lead Data Scientist who must handle the messy, chaotic edge cases of the real world.

- **Edge Case Focus:** Real data is rarely perfect. Expect zeros that mean "missing", outliers that are actually signals, and distributions that break standard assumptions.
- **Mathematical Rigor:** You must prove your intuition with formulas.
- **Algorithmic Depth:** Think about time complexity and geometric implications.

1 Problem Set A: The Archeologist's Dilemma (Data Preprocessing)

Context: The Lost City of Z

You are the lead data scientist for an archaeological expedition. You have discovered a clay tablet containing census data from an ancient civilization (3000 BC).

- **Dataset:** 10,000 rows representing citizens.
- **Features:** Age, Income_Grain_Sacks, Family_Size, Distance_to_River.
- **Edge Case 1 (The Zero Trap):** The variable Income has many zeros.
- **Edge Case 2 (The River Bias):** 15% of Income is missing (NaN), but only for citizens living far from the river.

Part 1: The Zero vs. Null Paradox (Real-World Logic)

1. **The Zero Ambiguity:** In real-world data, '0' can mean "No Income" (Real Value) or "Not Recorded" (Missing Value). How would you mathematically or statistically distinguish between these two cases without asking the dead citizens? (Hint: Look at distributions or correlations with Age). 2. **Missingness Mechanism:** Classify the missingness of Income (MCAR, MAR, MNAR) based on the "River Bias". Justify your answer using the definition of conditional probability $P(M|X)$.

Part 2: Mathematical Evaluation (Derivation)

Proof: The Flaw of Mean Imputation

Let X be the random variable for Income. The true population mean is μ . We split the population into two groups: Group A (Near River, observed) and Group B (Far River, missing). Assume $E[X_A] > E[X_B]$ (People near the river are richer). 1. Derive the mathematical bias term if we impute the missing values in Group B using the mean of Group A ($\bar{x}_A$).

$$\text{Bias} = E[\hat{\mu}_{\text{imputed}}] - \mu$$

Show that the bias is non-zero and positive.

Part 3: Algorithmic Strategy

1. Why would **Mean Imputation** be catastrophic here? 2. Propose an **Iterative Imputer (MICE)** strategy. Explain step-by-step how the algorithm uses **Distance_to_River** to "learn" that far-away citizens likely have lower income, effectively correcting the bias derived in Part 2.

Part 4: Python Implementation (Competitive Style)

Write a Python function `impute_ancient_data(df)` that: 1. Detects the correlation between missingness in **Income** and the value of **Distance_to_River**. 2. Applies an **IterativeImputer (MICE)** to fill the gaps. 3. Returns the imputed dataframe.

```
1 def impute_ancient_data(df):
2     # Your logic here (Mental Draft or Pseudo-code)
3     pass
```

2 Problem Set B: The Interstellar Navigator (Scaling)

Context: Warp Drive Coordinates

Your spaceship uses KNN to optimize fuel.

- X_1 : **Warp Factor** (Range: 0.0 to 9.9)
- X_2 : **Distance** (Range: 0 to 1,000,000,000 km)

Part 1: The Geometric Distortion

1. Calculate the Euclidean Distance between $A = (1.0, 100)$ and $B = (1.1, 1000)$. 2. **Mathematical Proof:** Show that as $X_2 \rightarrow \infty$, the contribution of X_1 to the distance metric approaches zero.

$$\lim_{X_2 \rightarrow \infty} \frac{\partial(\text{Distance})}{\partial X_1} \approx 0$$

Part 2: Edge Case - The "Asteroid" Outlier

A sensor glitch creates one data point: $X_{\text{glitch}} = (5.0, 10^{15})$. 1. If you apply **Min-Max Scaling**, what is the new range of the *valid* data points? Show that they are crushed into an interval $[0, \epsilon]$ where $\epsilon \approx 10^{-6}$. 2. Why does this "crushing" destroy the gradient descent process in Neural Networks? (Think about the derivative of the Sigmoid/ReLU function on very small numbers).

Part 3: Implementation

Write a Python snippet using `sklearn` to preprocess this data, ensuring the "Asteroid Glitch" does not ruin the model.

3 Problem Set C: The Needle in the Haystack (Imbalance)

Context: Rare Disease Detection

- **Population:** 1,000,000 patients.
- **Positives (Sick):** 100 (0.01%).
- **Edge Case (The Cost of Life):** A False Negative (missing a sick person) costs \$1,000,000 (Lawsuit). A False Positive (unnecessary test) costs \$100.

Part 1: The Cost Function (Mathematical Abstraction)

1. Define a custom Loss Function $L(y, \hat{y})$ that incorporates the "Cost of Life".

$$L = \alpha \cdot FN + \beta \cdot FP$$

Calculate the ratio α/β . 2. Prove why optimizing for standard **Accuracy** minimizes the wrong loss function in this scenario.

Part 2: SMOTE Geometry

1. **Interpolation Logic:** Explain geometrically how SMOTE generates a new point $\vec{x}_{new}$ between a minority point $\vec{x}_i$ and its neighbor $\vec{x}_j$.

$$\vec{x}_{new} = \vec{x}_i + \lambda(\vec{x}_j - \vec{x}_i)$$

What is the domain of λ ? 2. **Edge Case Risk:** What happens if $\vec{x}_i$ is an outlier (noise) located deep inside the majority cluster? Does SMOTE help or hurt here?

Part 3: Competitive Coding

Write a Python block that: 1. Splits the data into Train/Test. 2. Applies SMOTE **only** to the Training set. 3. Trains a Random Forest classifier.

4 Problem Set D: The Alien Signal (Unsupervised Learning)

Context: SETI Signal Processing

You receive radio signals.

- **Cluster A:** A dense, spherical core.
- **Cluster B:** A sparse "Ring" surrounding Cluster A.
- **Edge Case (Noise):** Random background radiation scattered everywhere.

Part 1: Mathematical Evaluation of K-Means

Proof: The Spherical Assumption

The K-Means objective function minimizes SSE: $J = \sum ||x_i - \mu_k||^2$. 1. Prove that the set of points $\{x\}$ closer to μ_1 than to μ_2 forms a convex set (specifically, a half-space separated by a hyperplane).

$$||x - \mu_1||^2 < ||x - \mu_2||^2 \iff w^T x + b > 0$$

2. Use this linear separation property to explain **why** K-Means cannot separate the "Ring" from the "Core" (since a ring is non-convex).

Part 2: DBSCAN Logic

1. How does the density-reachability condition of DBSCAN allow it to "chain" together the points in the Ring cluster? 2. **Hyperparameter Edge Case:** What happens to the clusters if the background radiation (noise) becomes slightly denser than the Ring? (Think about ϵ and MinPts).

Part 3: Stratified Semi-Supervised Learning

You have budget to label 50 signals. 1. **Algorithm Design:** Design a pseudo-code algorithm that uses K-Means to select the 50 most *representative* samples (closest to centroids) to label manually. 2. **Constraint:** Why must we use a large K (e.g., $K = 50$) for this sampling step, even if the true number of clusters is only 3?